Point cloning and the gonality-two speculation

This is a separate question-track candidate, not part of the principal-source credit for L009. It has not yet received independent review. Graph blow-ups and twin fibers are standard constructions; no priority is asserted for them or for this application.

Literal source scope

The final journal article of Delaby–Leemans–Tranchida, *Geometries with trialities arising from linear spaces*, Algebraic Combinatorics 9(2) (2026), 379–402, says on printed page401, after specifying input gonality two: "However, experiments suggest that $\Delta(\Gamma)$ will not be residually connected in that case." This is an experimental suggestion, not a named or numbered conjecture. The bound journal PDF and its hash are recorded in `phase265_intake.md`.

The input of Construction1 is a rank-two geometry under the definitions on pp381–383. Distinct objects of the same type are not incident. Every maximal flag must be a chamber. The definitions do not require distinct objects to have different neighborhoods, nor do they require the input to be a linear space. The incidence graph is a simple bipartite graph; simplicity does not exclude twin vertices. The example below deliberately has twins. It makes no statement about the additional problem in which one requires distinct point neighborhoods, or about the separate gonality-three nonlinear question.

The cloning lemma

Let $S = (P, \mathcal{L}, \in)$ be a rank-two geometry and let $a \geq 2$. Replace every point p by a distinct points (p, c), $1 \leq c \leq a$, and retain every line. Put $(p, c) \in L$ if and only if $p \in L$. Denote the resulting geometry by $S^{[a]}$. It is still a rank-two geometry: every point and line lies on an incidence, and its maximal flags are precisely the incident point-line pairs. If some point of S lies on two distinct lines, its two clones and those lines give a four-cycle. There are no loops or parallel edges, so the input has gonality exactly two.

The triangle geometry $\Delta(S^{[a]})$ is exactly the independent vertex blow-up of $\Delta(S)$, replacing each vertex by an independent set of size a. Indeed, a flag of $S^{[a]}$ is (p, c, L), and the two conditions in Construction1 are

$$p \in L', \qquad p' \notin L,$$

independent of both clone coordinates. Same-type distinct vertices remain nonincident, as required.

More generally, let X be a finite-rank incidence geometry and replace every vertex by a clones of the same type, pulling back every cross-type incidence. Every flag projects to a flag of X and extends to a chamber by lifting a base chamber, so the blow-up is a geometry. The residue of any lifted flag is exactly the blow-up of the residue of its projected flag: the remaining types are unchanged, and their incidences are pulled back. Every panel size is multiplied by a.

If X is residually connected, so is the blow-up. A connected residue of rank at least two has no isolated vertex, since every vertex extends to a chamber. A path between distinct base vertices lifts to a path between any selected clones of its endpoints. Two clones of the same base vertex can be connected through a clone of a neighboring base vertex. This proves connectivity of every rank-at-least-two residue, including the whole geometry. Firmness is likewise preserved.

If X is flag-transitive, so is its blow-up. Lift the base automorphisms by leaving clone labels fixed, and allow independent permutations of the clones over every base vertex. These are incidence automorphisms.

First map the projections of two chambers by a base automorphism, then adjust the clone label separately in each of their distinct type fibers. The resulting automorphism sends one lifted chamber to the other.

A literal finite counterexample

Take $S = \mathrm{AG}(2, 3)$ and $a = 2$. The input $S^{[2]}$ has eighteen points, twelve lines and seventy-two incidences. Each point lies on four lines and each line contains six points. Its graph is simple, connected and has girth four.

The base triangle geometry $\Delta(S)$ has thirty-six vertices per type and 432 chambers. Its rank-two residues are twelve-cycles, and it is flag-transitive under the affine group of order432. These facts have complete proofs in the retained incidence argument. The accompanying program independently constructs the cloned geometry directly from the nine-point coordinate plane and checks its asserted properties. The affine group is transitive on ordered noncollinear triples because its linear part is transitive on ordered bases.

The cloning lemma therefore makes $\Delta(S^{[2]})$ a firm, residually connected, flag-transitive geometry with seventy-two vertices per type and $432 \cdot 2^3 = 3456$ chambers. Every panel has size four. Every rank-two residue has twenty-four vertices and degree four. This gives a counterexample to the unrestricted gonality-two suggestion using the literal source definitions. It does not claim a counterexample under a supplementary no-twins hypothesis.

The independent-vertex blow-up viewpoint also gives an infinite family: every integer $a \geq 2$ yields a gonality-two input and preserves the three properties of its triangle geometry. The proof of this quantifier is the cloning lemma; no extrapolation from the case $a = 2$ is used.

The program `phase265_gonality_two_check.py` constructs the input from coordinate incidences, implements the source triangle relation directly, enumerates all chambers and panel sizes, checks the whole graph and every rank-two residue, and checks a transitive chamber action by actual permutations. It does not use a precomputed blow-up adjacency matrix as its definition. No full Lean formalization or present-literature priority determination is claimed.